

Zac Stewart

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Professional Summary

Computer Vision Engineer with 16 years of experience delivering quantifiable business impact, from startups to Fortune 500. Seeking new opportunities to pioneer solutions using computer vision and machine learning.

Experience

Freelance Computer Vision & ML Engineer, 2015 - Present

Agentic Personal Finance Assistant, [Frank](#)

- Helped lay the groundwork for SmartPath's new AI-powered personal finance assistant, Frank
- Built out an array of tools and sub-agents to enable Frank to answer questions like "Am I saving enough for retirement?"
- Structured agents to use deterministic tools like calculators instead of non-deterministic token prediction when doing things like calculating future investment values
- Tech/skills: Python, agents, LLMs

Collectable Trading Card AI Grading, [chck.ai](#), formerly Shoebox

- Helped the client launch an AI-powered iPhone app to scan and grade the condition of collectable trading cards
- Built a camera calibration system using differentiable to collate 2D damage detections from 52 camera views into a single 3D coordinate space
- Designed a dependency graph to coordinate all subtasks involved with grading (semantic segmentation, object detection, and more)
- Built a resource allocation system to launch and scale components as needed, and saved this startup thousands of dollars per month in compute bills
- Tech/skills: Python, OpenCV, PyTorch, PyTorch3D, object detection, semantic segmentation, differentiable rendering, camera calibration, MLOps, AWS

3D Markerless Motion Capture System for Professional Soccer Leagues, [S3Global](#)

- Helped the client win a live TV broadcast contract by inventing an algorithm to triangulate 3D positions of 22 players and the ball simultaneously based on frame-synchronized 2D object detections from 20 cameras in real time (120+ FPS).
- Enabled all 3D computer vision research and development to proceed by creating a process to calibrate 20 cameras using the low-feature, high-symmetry imagery from installations in soccer stadiums.
- Halved 3D triangulation reprojection error by developing a process to refine camera calibration using feedback from our own 3D object triangulations.
- Tech/skills: Python, C++, OpenCV, COLMAP, epipolar geometry (geometry involving two perspectives of a 3D scene), multi-view stereo computer vision, camera calibration, RANSAC, Redis

AI-Powered Video Auto-Cropping, [Munch](#)

- Reduced customer churn by developing and implementing a custom algorithm for video aspect ratio reformatting, outperforming the previous vendor-provided Cloudinary service with higher-quality AI-cropped videos in challenging scenarios.
- Reduced operational cost by more than 10x, and processing time by 6x.
- Tech/skills: Python, OpenCV, object detection, object tracking, video rendering, FFmpeg

Retrieval Augmented Generation (RAG) Q&A Chatbot, Confidential Publisher

- Collaborated with the client to launch a cutting-edge generative AI chatbot, providing a new way for readers to interact with their extensive catalog of content, and increasing customer engagement.
- Led the design and development of catalog ingestion pipeline and semantic search engine, integrating with a large language model (LLM).
- Empowered editorial team to shape the chatbot's writing style and tone by building tooling for prompt engineering and tracing semantic search results used to generate responses.
- Tech/skills: Python, pgvector, text embedding models, AWS Bedrock

Personalized 3D Avatar Generator, Confidential Startup

- Collaborated with the client to develop a toolkit to create personalized 3D avatars from a face scans.
- Created a program that captured high-speed RGB and depth frames, alongside audio, which was used to scan models performing various "emotes" with an Intel RealSense 3D camera (similar to Xbox Kinect).
- Developed a program that stitched together depth frames from a capture to produce a 3D mesh of the model.
- This startup and the IP I created for them was acquired by a major gaming company before making it to the market.
- Tech/skills: Python, C++, Point Cloud Library (PCL), 3D depth imagery, iterative closest point (ICP) algorithm

Maintenance Work Order Multi-Class Message Routing System, [Dude Solutions \(now Brightly, A Siemens Company\)](#)

- Collaborated directly with the Chief Strategy Officer to design and pilot a new product used by building managers to automatically route maintenance requests to one of 11 departments.
- Achieved 92% validation accuracy by designing, training, and deploying a machine learning model that utilized natural language processing in tandem with custom feature engineering.
- Enabled a pilot with client's existing customers by deploying the ML model to production as a simple web service that was integrated into the client's

existing product offering.

- The pilot succeeded, saving their customers time and frustration creating maintenance requests. The company and the IP I created for them was later acquired by Siemens.
- Tech/skills: Python, scikit-learn, NLP, feature engineering, Flask

Lead Software Engineer (LMTS), Salesforce, Atlanta, GA, October 2017 - November 2020

- Vastly improved our customers' email engagement metrics by researching, designing, and leading development of an innovative system to detect and ignore email open and link click events originating from email security scanners like Barracuda.
- Worked cross-functionally with leads from other teams to collect traffic logs and estimate the effects of security scanner traffic on features under their responsibility. Took advantage of an internal hackathon to model the problem and prove that we could statistically identify and ignore the traffic. Pitched a working prototype to EVP-level management who green-lighted committing my team to the project for two release cycles.
- Brought the security scanner traffic filter service from prototype to general availability over the next six months by designing core pieces and delegating responsibilities to developers on my team. Held weekly meetings with leads from other teams to collaborate on impacts it would have on their product areas, as well as garner support from other teams whose infrastructure we relied upon.
- Salesforce was assigned a patent for the security scanner traffic filter listing me as the sole inventor ([US11552896B2: Filtering network traffic from automated scanners](#)).
- Closed numerous cross-site scripting security vulnerabilities at once by designing and implementing a framework to sandbox numerous WYSIWYG editors that my team was responsible for.
- Released a new template language for our email pipeline as part of a team. My most impactful contributions were extensive profiling and performance testing which revealed that our initial implementation would cause untenable email sending latency. Thereafter I led an effort over several sprints to optimize performance, and specifically implemented a Least Recently Used (LRU) cache to avoid redundantly parsing template fragments.
- I was promoted from Senior Engineer to Lead Engineer, reflecting my ability to work cross-functionally with other teams, present to diverse audiences including executives, lead my team in accomplishing release goals, my strong technical knowledge, and ability to plan and execute.
- Mentored two engineers one level below me as part of an internal mentorship program, helping them to earn promotions by setting goals, monitoring their progress, talking about perceived barriers and how to break through them.
- Tech/skills: Go, gRPC, Protocol Buffers, Python, Jupyter, Kubernetes, Graphite, Prometheus, Grafana, JavaScript, MySQL, time series analysis

Lead Developer, DirtJockey, Atlanta, GA, September 2015 – July 2017

- Built a specialized CRM that allowed heavy equipment dealers to quickly and intuitively update their inventory, keep track of leads and quotes, publish to advertising sites, and manage their own website.
- Enabled our customers to close deals and sell machines by fielding support calls and working with customers to gather product feedback, which we then incorporated into product design and development. Tech/skills: Ruby, Ruby on Rails, PostgreSQL, Ember

Software Developer & Instructor, Big Nerd Ranch, Atlanta, GA, August 2012 - February 2015

- Co-authored a course and guidebook on Ruby web development.
- Taught several week-long, intensive boot camps using the course I co-authored. Received great reviews from students, saying that the course helped them grow in their career path.
- Worked on numerous client projects as a backend software engineer with many happy clients.
- Built a computer vision iOS app to digitize crossword puzzles in a 48-hour internal hackathon.
- Organized the 2013 [Space Apps Atlanta](#) hackathon.
- Gave several talks at local meetups, including PyATL and the Atlanta Ruby Users group.
- Tech/skills: Ruby, Ruby on Rails, Sinatra, DocBook, Objective-C, iOS, OpenCV, d3.js

Software Engineer, Emcien, Atlanta, GA, July 2011 - July 2012

- Worked with a team to build a Rails backend to expose a patented pattern engine as a SaaS.
- Enabled customers to understand complex graph relationships discovered by our pattern analysis engine by creating intuitive visualizations.
- Enabled the business to close a deal that involved larger datasets than we'd previously worked with by working non-stop to design and implement a caching scheme during a short period of time.
- Was promoted from Junior Software Engineer to Software Engineer, reflecting my ability to work independently, lead important initiatives, and reliably hit deadlines. Tech/skills: Ruby, Ruby on Rails, d3.js, InfoVis, Raphaël, Highcharts, Backbone.js